



AIM – 9933
BUOYANCY BALANCE



Aimil Ltd.
Instrumentation & Technologies

Version

2.0



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BUOYANCY BALANCE

Operating Instructions



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Buoyancy Balance

Introduction

This equipment is used for determining the density of hardened concrete in the following conditions: -

- a. As received in the Lab.
- b. Saturated
- c. Owendried

Description

This equipment is illustrated in the General Assembly Drawing and the numbers given against the components of the equipment in the description below pertain to the General Assembly Drawing.

It consists of Electronic Balance (5) having a capacity of 15kg and equipped with a stirrup for weighing the specimen in both air and water to an accuracy of 0.5gm. The balance which operates on a 220V, Single Phase power supply is mounted on a Rigid support frame (1). The Water Tank of dimensions as 400L x 250W x 300H is on the base of the frame. The support frame is provided with a tank lifting device to raise the water tank while weighing the specimen in water (arrangement shown in enclosed GA drawing).

Operation

Fill the water tank upto about 2cms below the top edge of the Tank. Place the Electronic Digital Balance on top of the frame in such a way that the hole for the suspension hook (6) aligns with hole for the stirrup provided at the bottom of the balance. Suspend the hook in the stirrup. Switch the balance on, wait for a few seconds to enable the balance to stabilise. Set the reading to zero, by pressing the tare switch on the balance.

Place the specimen in the specimen holder (3), suspend it in the hook and make a note of reading. This is the weight of the specimen in air.

Raise the Water Tank by using the Mechanical Device (4) and totally immerse the specimen in water. Make a note of the reading. This is the weight of the specimen in water.



Precautions

Ensure that hook does not touch the bottom of the tank and that the air bubbles are not trapped on the specimen & the hook.

Calculation

Weigh the completely immersed specimen & record its apparent mass (M_w) in kg, after correcting for the apparent mass of the empty stirrup in water to the same depth as the specimen. Immediately remove the specimen from the stirrup & after wiping off the surplus water from the surfaces using a moist cloth, weigh in air & record its mass (M_a) in kg.

Calculate volume of the specimen (V) in cu-m, taking the density of water to be 1000kg/cu-m:

$$(V) = \frac{M_a - M_w}{1000}$$

A. Procedure for the Determination of As-Received Density

1. Weigh the specimen as - received in air & record its mass (m_1) in kg.
2. Calculate the as - received density using the appropriate formula :-

i. $P_1 = \frac{m_1}{V}$

Where V_1 is the volume of the cylindrical cored specimen in cu-m calculated by taking the average length of the diameter of the specimen.

ii. $P_2 = \frac{m_1}{V_2}$

Where V_2 is the volume of moulded specimens i.e, cubes & beams calculated by using calipers, in cu-m.

iii. $P_3 = \frac{m_1}{V}$

Where V is the volume obtained by the displacement of water, in cu-m.



B. Procedure for Determination of Saturated Density

1. Fully immerse the specimens in water at $20 \pm 2^\circ\text{C}$ until constant mass is achieved.
2. Before each weighing, wipe off the surplus water from the surface using a moist cloth.
3. Record the constant mass (m_2) in kg.

Calculation

Saturated densities are :-

1. $P_4 = \frac{m_2}{V_1}$ Where V_1, V_2, V & M_2 are as mentioned in A.

2. $P_5 = \frac{m_2}{V_2}$

3. $P_6 = \frac{m_2}{V}$

C. Procedure for Determination of Oven Dried Density

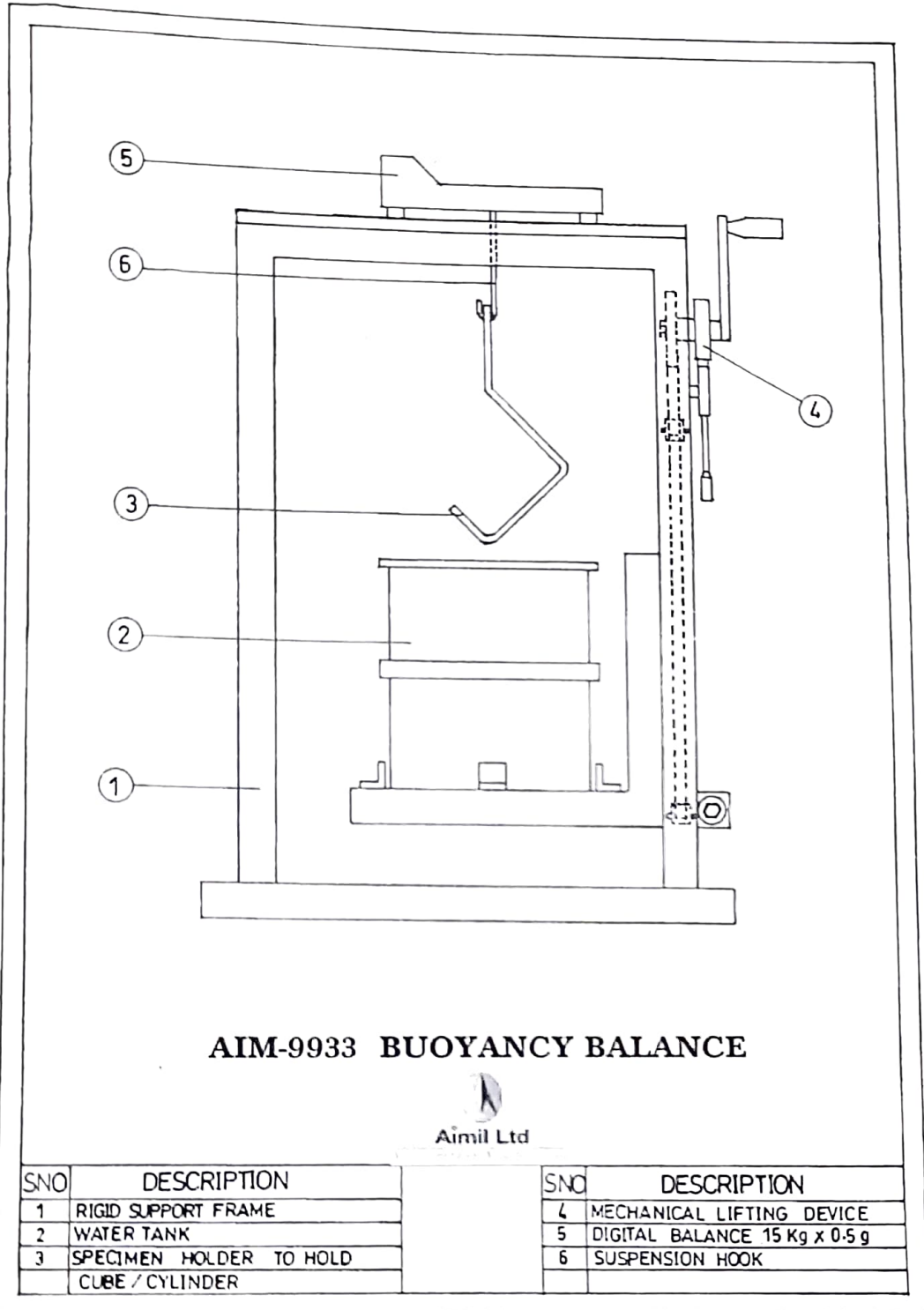
1. Place the specimens in the ventilated Oven & dry it at $105 \pm 5^\circ\text{C}$ until a constant mass is achieved.
2. Before each weighing, cool the specimen to near room temperature in a dry air tight vessel or desiccator.
3. Record the constant mass (m_3) in kg.

Over dried densities are calculated as :-

i. $P_7 = \frac{m_3}{V_1}$ Where V_1, V_2, V_3 are as mentioned in A.

ii. $P_8 = \frac{m_3}{V_2}$

iii. $P_9 = \frac{m_3}{V_3}$



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SNO	DESCRIPTION
1	RIGID SUPPORT FRAME
2	WATER TANK
3	SPECIMEN HOLDER TO HOLD CUBE / CYLINDER

SNO	DESCRIPTION
4	MECHANICAL LIFTING DEVICE
5	DIGITAL BALANCE 15 Kg x 0.5 g
6	SUSPENSION HOOK